

Homework 3

DATA 624

Kevin Havis

September 19, 2025

Table of contents

1 Homework 3	1
1.1 Exercise 5.1	1
1.2 Exercise 5.2	11
1.3 Exercise 5.3	15
1.4 Exercise 5.4	17
1.4.1 Australian Exports	17
1.4.2 Bricks	19
1.5 Exercise 5.7	20

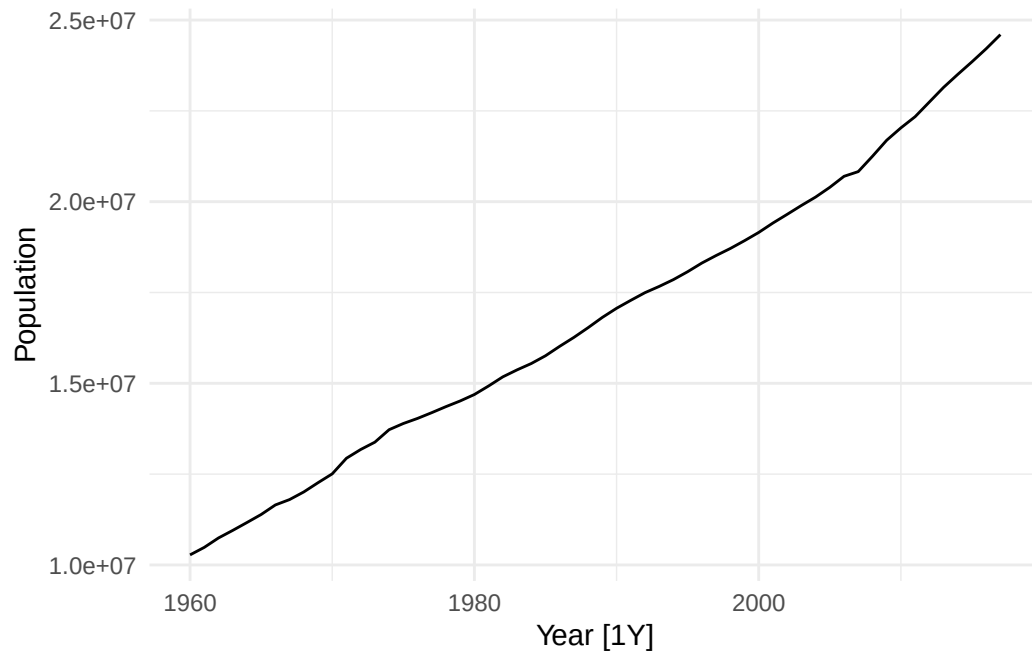
1 Homework 3

1.1 Exercise 5.1

Produce forecasts for the following series using whichever of `NAIVE(y)`, `SNAIVE(y)` or `RW(y ~ drift())` is more appropriate in each case: * Australian Population (`global_economy`) * Bricks (`aus_production`) * NSW Lambs (`aus_livestock`) * Household wealth (`hh_budget`) * Australian takeaway food turnover (`aus_retail`)

```
# Australian Population
ap <- tsibbledata::global_economy |>
  filter(Country == "Australia") |>
  select(Year, Population)

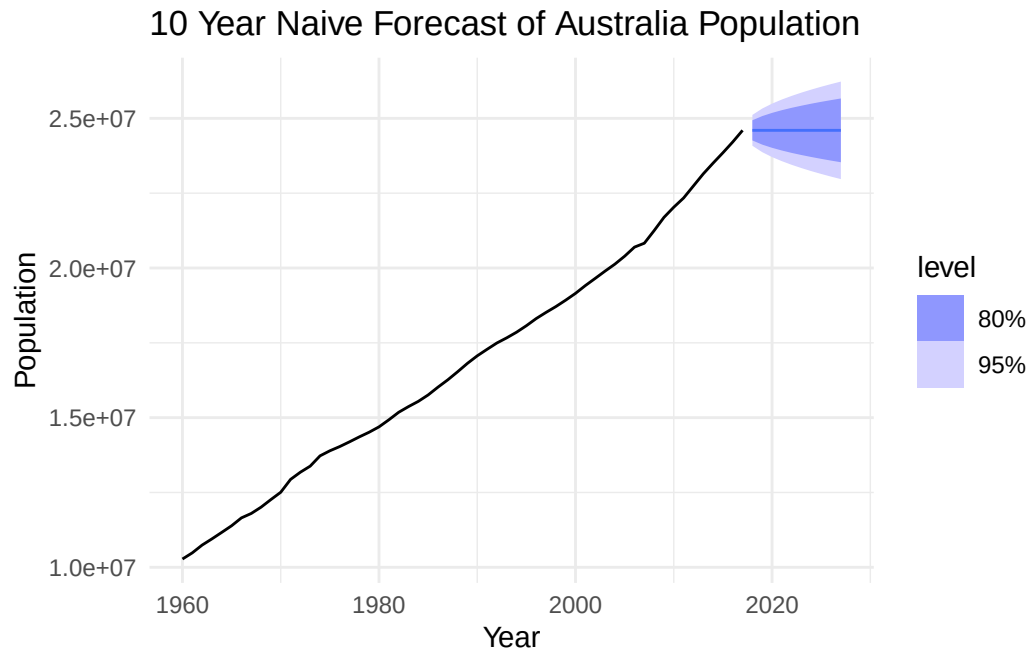
# Visualize to pick model
autoplot(ap)
```



```
# We choose Naive due to straightforward linear data
ap_naive <- ap |> model(NAIVE(Population))

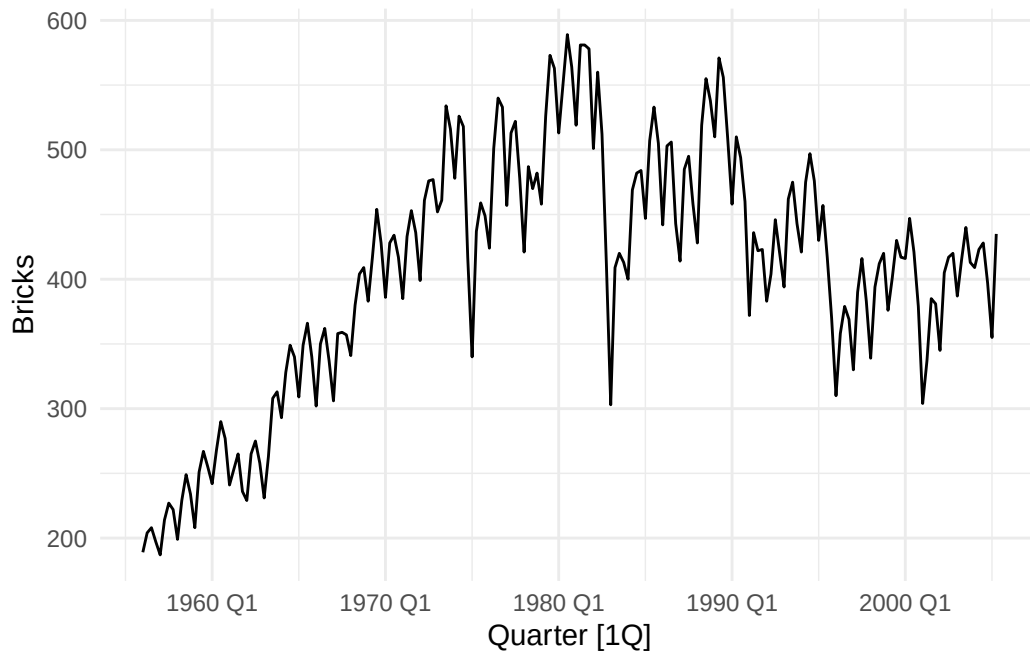
ap_fc <- ap_naive |> forecast(h=10)

ap_fc |> autoplot() + autolayer(ap) +
  labs(title="10 Year Naive Forecast of Australia Population") +
  guides(color = guide_legend(title="Forecast"))
```



```
# Bricks
bricks <- tsibbledata::aus_production |>
  select(c(Quarter, Bricks)) |>
  filter(!is.na(Bricks))

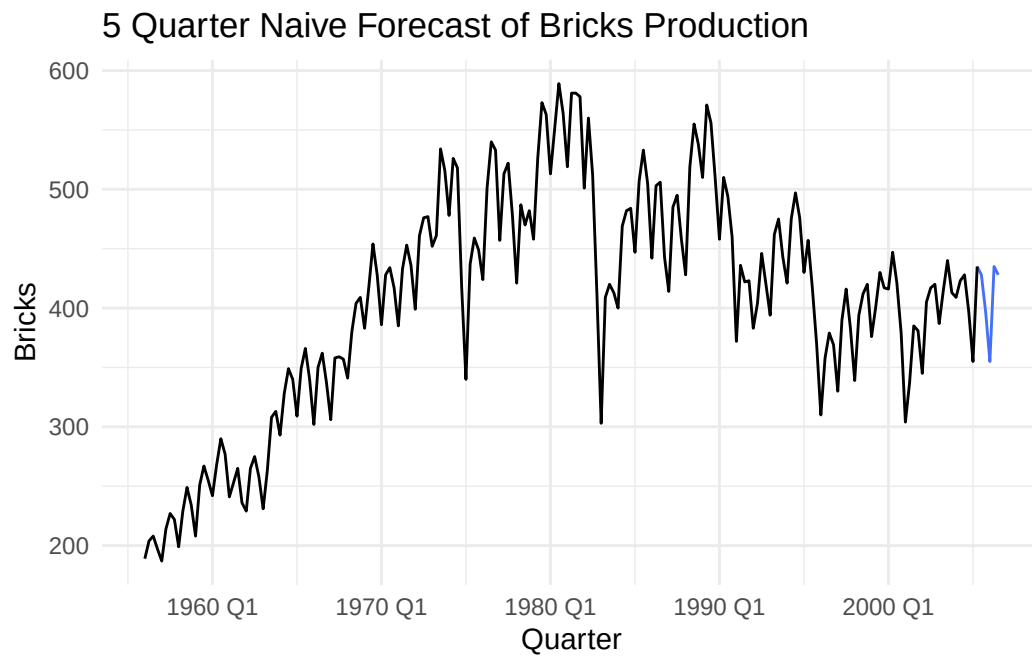
# Visualize to pick model
autoplot(bricks)
```



```
# We choose Naive due to straightforward linear data
bricks_snaive <- bricks |> model(SNAIVE(Bricks))

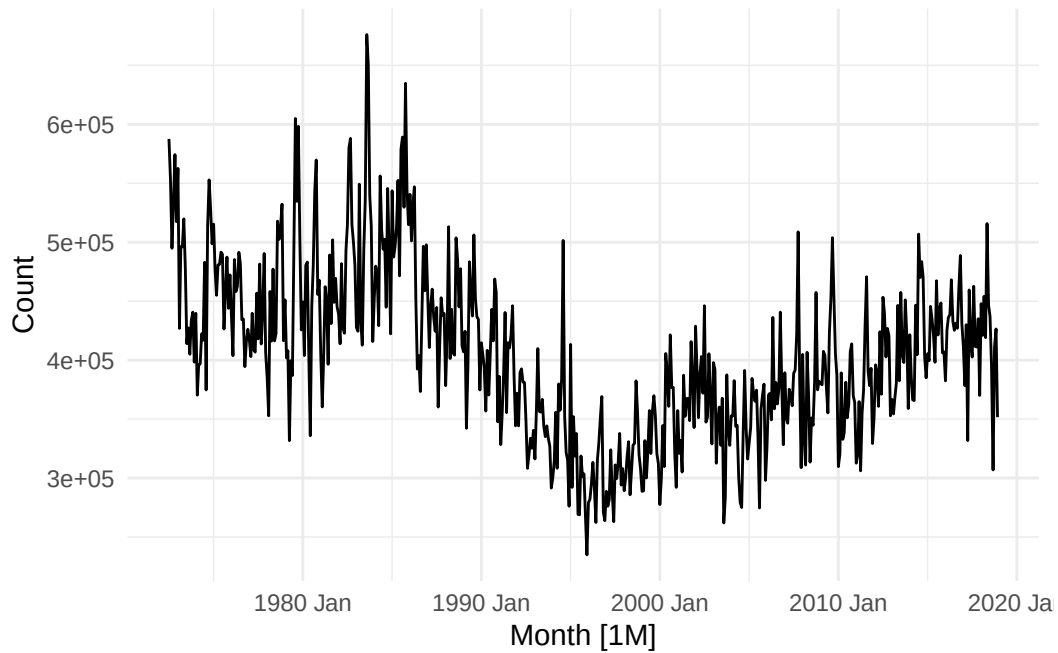
bricks_fc <- bricks_snaive |> forecast(h=5)

bricks_fc |> autoplot(bricks, show_gap = FALSE, level = NULL) +
  labs(title = "5 Quarter Naive Forecast of Bricks Production") +
  guides(color = guide_legend(title = "Forecast"))
```



```
# NSW Lambs
nsw <- tsibbledata::aus_livestock |>
  filter(Animal == "Lambs", State == "New South Wales") |>
  filter(!is.na(Count))

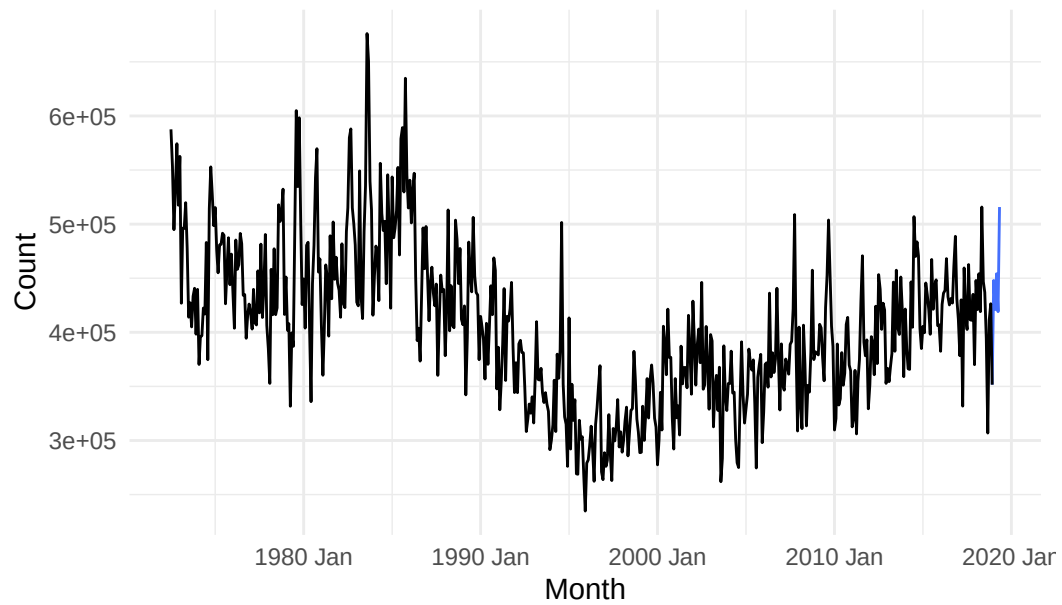
# Visualize to pick model
autoplot(nsw)
```



```
# We choose SNAIVE due to seasonal patterns in livestock data
nsw_snaive <- nsw |> model(SNAIVE(Count))
nsw_fc <- nsw_snaive |> forecast(h = 5)

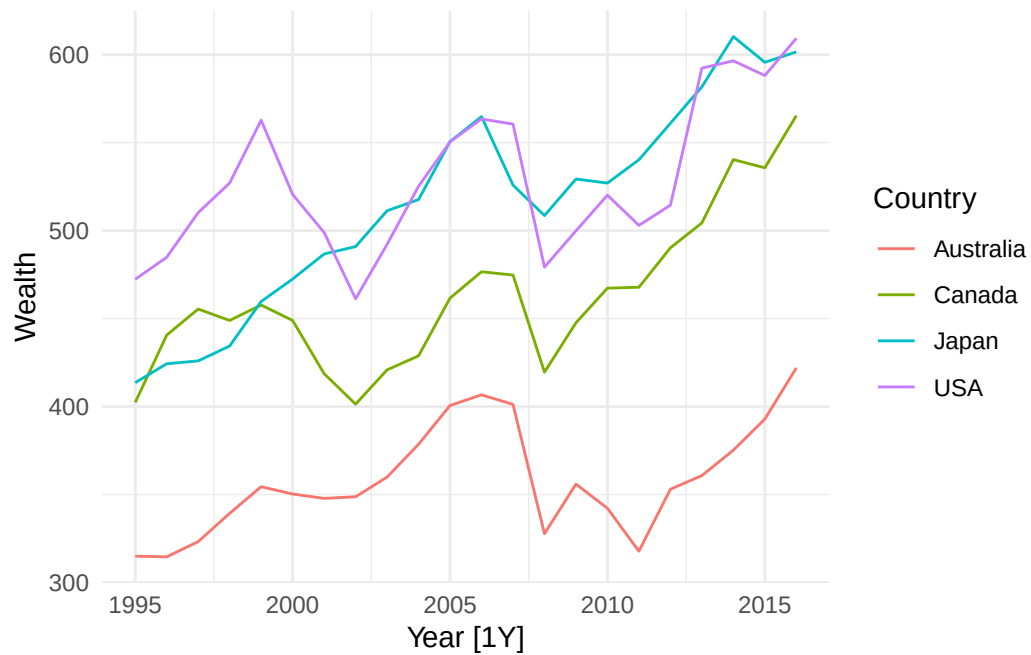
nsw_fc |> autoplot(nsw, show_gap = FALSE, level = NULL) +
  labs(title = "5 Period Seasonal Naive Forecast of NSW Lambs") +
  guides(color = guide_legend(title = "Forecast"))
```

5 Period Seasonal Naive Forecast of NSW Lambs



```
# Household wealth
wealth <- tsibbledata::hh_budget |>
  select(Year, Wealth) |>
  filter(!is.na(Wealth))

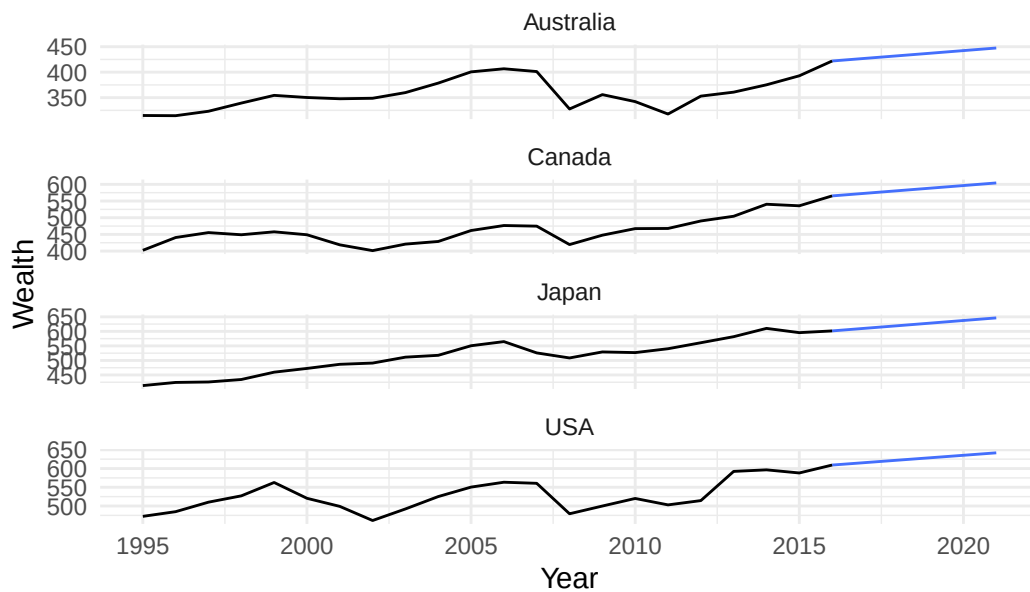
# Visualize to pick model
autoplot(wealth)
```



```
# We choose Naive with drift due to trending data
wealth_naive <- wealth |> model(NAIVE(Wealth ~ drift()))
wealth_fc <- wealth_naive |> forecast(h = 5)

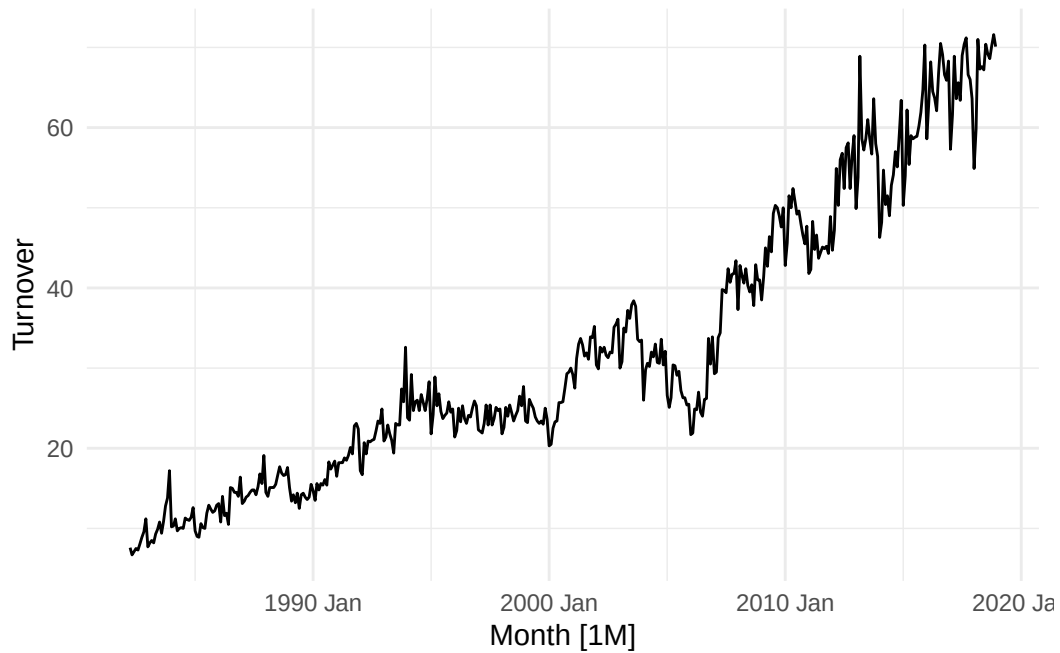
wealth_fc |> autoplot(wealth, show_gap = FALSE, level = NULL) +
  labs(title = "5 Year Naive with Drift Forecast of Household Wealth") +
  guides(color = guide_legend(title = "Forecast"))
```

5 Year Naive with Drift Forecast of Household Wealth



```
# Australian takeaway food turnover#
# Filtering for Australian Capital Territory for simplicity
takeaway <- tsibbledata::aus_retail |>
  filter(Industry == "Cafes, restaurants and takeaway food services") |>
  filter(!is.na(Turnover)) |>
  filter(State == "Australian Capital Territory")

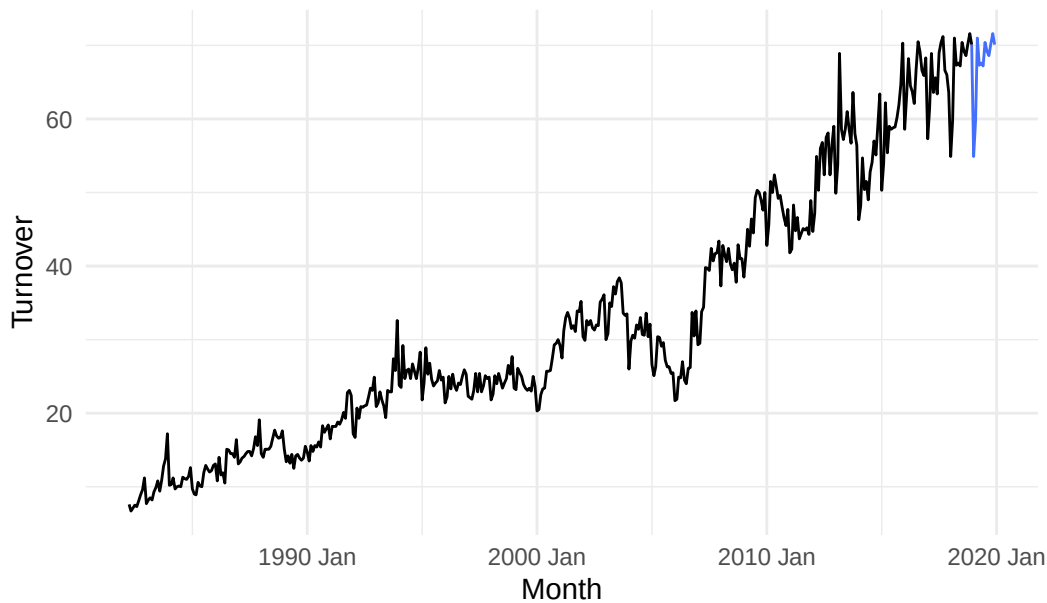
# Visualize to pick model
autoplot(takeaway)
```



```
# We choose SNAIVE due to seasonal patterns in retail data
takeaway_snaive <- takeaway |> model(SNAIVE(Turnover))
takeaway_fc <- takeaway_snaive |> forecast(h = 12)

takeaway_fc |> autoplot(takeaway, show_gap = FALSE, level = NULL) +
  labs(title = "12 Period Seasonal Naive Forecast of Takeaway Food Turnover") +
  guides(color = guide_legend(title = "Forecast"))
```

12 Period Seasonal Naive Forecast of Takeaway Food Turnover



1.2 Exercise 5.2

Use the Facebook stock price (data set `gafa_stock`) to do the following: 1. Produce a time plot of the series. 2. Produce forecasts using the drift method and plot them. 3. Show that the forecasts are identical to extending the line drawn between the first and last observations. 4. Try using some of the other benchmark functions to forecast the same data set. Which do you think is best? Why?

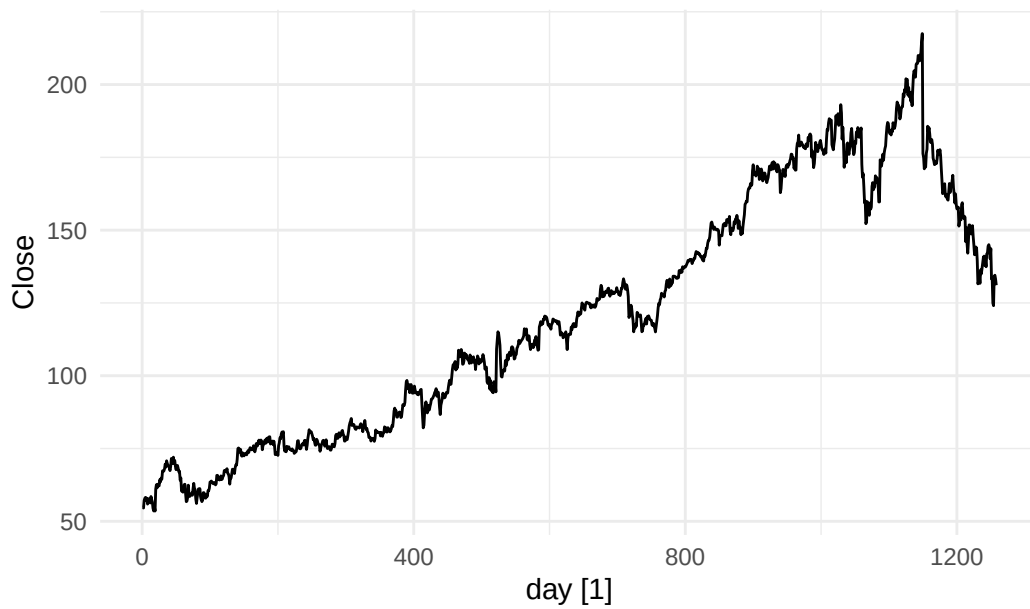
In this case I would prefer the naive model. The drift model is correctly picking up the trend of the overall timeseries, however given the sudden peak and descent of the data, I think it is erroneous to assume the previous upward trend is still in effect.

The naive model will give us a simple estimate, but in this scenario, a more reasonable one.

```
# Produce time plot
fb <- tsibbledata::gafa_stock |>
  filter(Symbol == "FB") |>
  mutate(day = row_number()) |>
  select(day, Close) |>
  update_tsibble(index = day, regular = TRUE)

autoplot(fb)+labs(title="Facebook Share Price")
```

Facebook Share Price

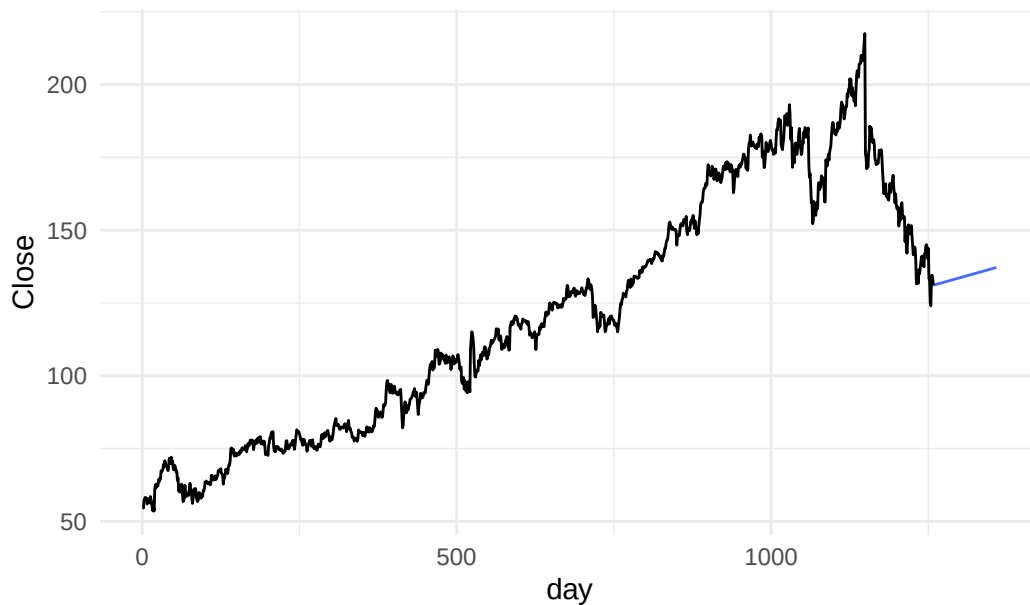


```
# Produce forecasts
fb_drift <- fb |>
  model(NAIVE(Close~drift()))

fb_fc <- fb_drift |> forecast(h=100)

fb_fc |> autoplot(fb, show_gap=FALSE, level=NULL)+labs(title="Drift model forecasting FB Share Price")
```

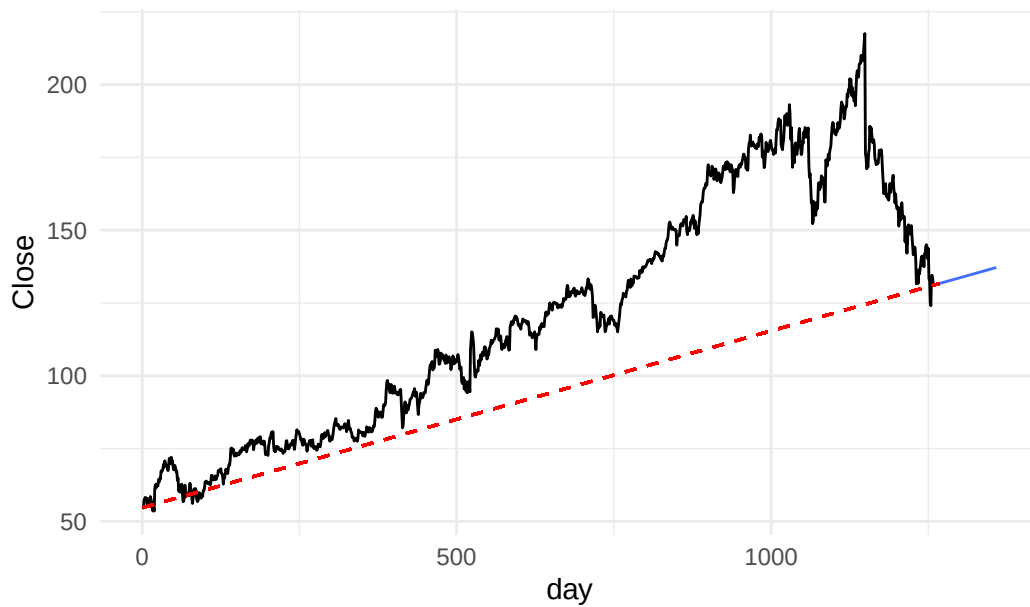
Drift model forecasting FB Share Price



```
# Manually calculate drift slope with first and last obs
first_obs <- fb |> slice_head(n=1)
last_obs <- fb |> slice_tail(n=1)

fb_fc |> autoplot(fb, show_gap = FALSE, level = NULL) +
  geom_segment(aes(x = first_obs$day, y = first_obs$Close,
                  xend = last_obs$day + 10, yend = last_obs$Close +
                    ((last_obs$Close - first_obs$Close) / (last_obs$day - first_obs$day)) * 10,
                  color = "red", linetype = "dashed", size = 0.5) +
  labs(title="Drift Model vs First and Last Observation Vector")
```

Drift Model vs First and Last Observation Vector

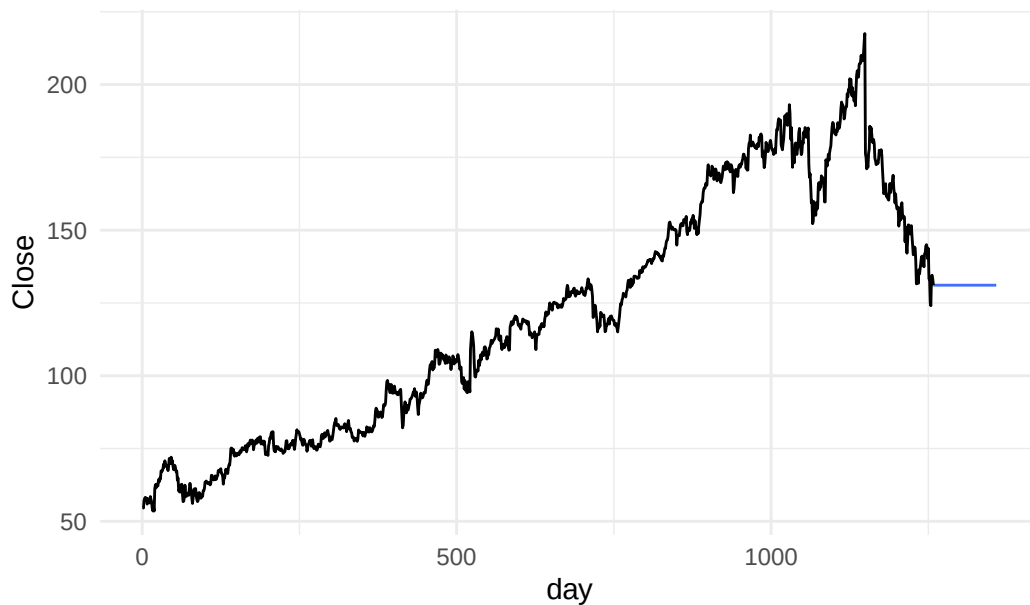


```
fb_naive <- fb |>
  model(NAIVE(Close))

fb_naive_fc <- fb_naive |> forecast(h=100)

fb_naive_fc |> autoplot(fb, show_gap=FALSE, level=NULL)+labs(title="Naive Forecast of Facebook")
```

Naive Forecast of Facebook Share Price



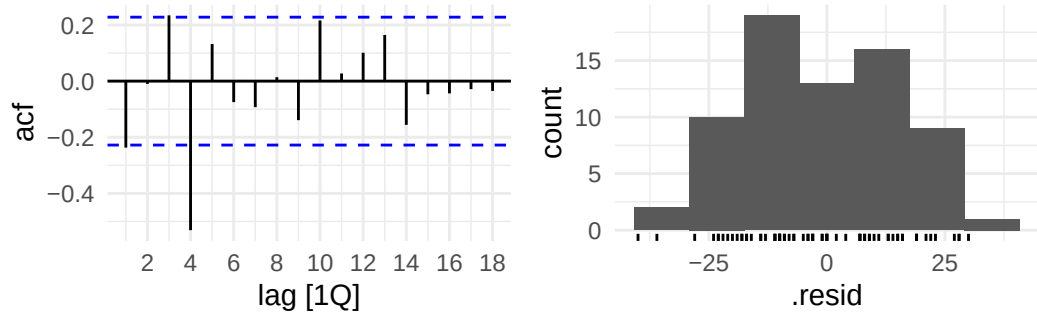
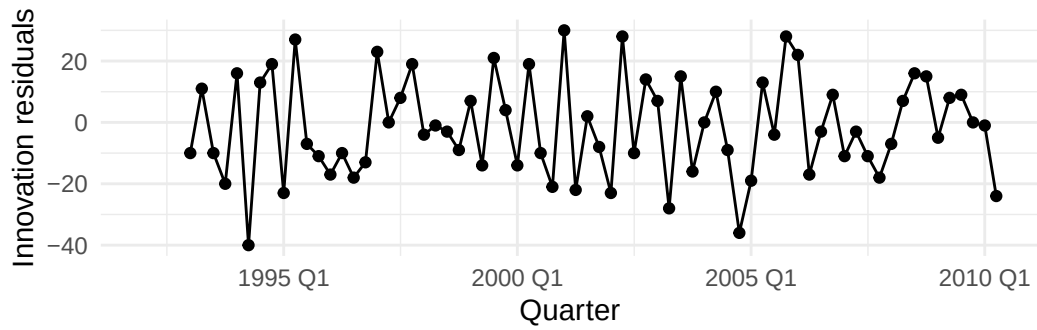
1.3 Exercise 5.3

Apply a seasonal naïve method to the quarterly Australian beer production data from 1992. Check if the residuals look like white noise, and plot the forecasts. The following code will help.

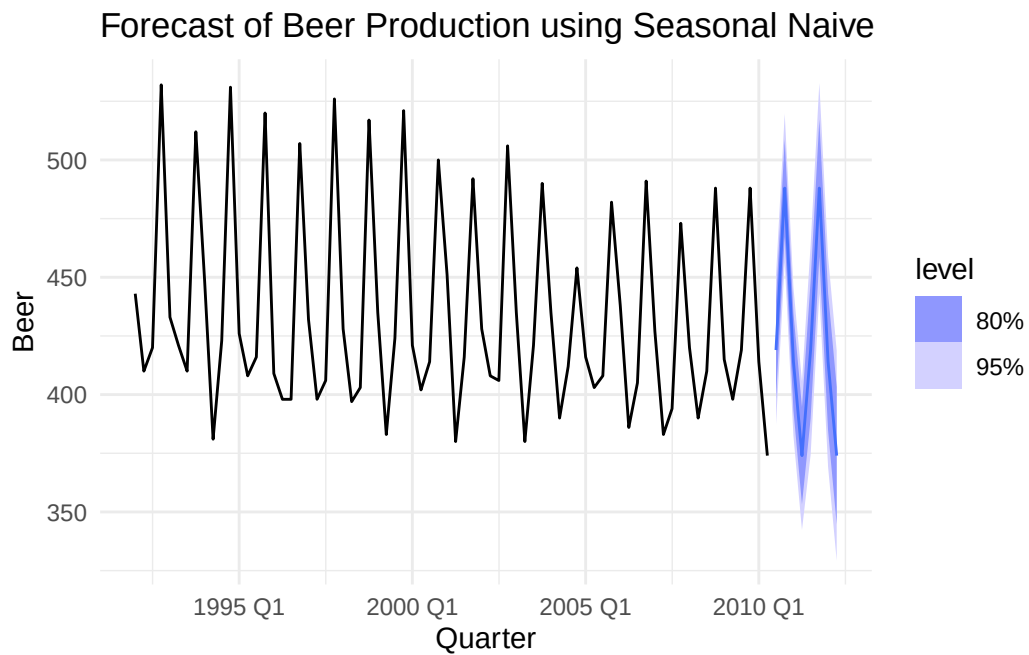
```
# Extract data of interest
recent_production <- aus_production |>
  filter(year(Quarter) >= 1992)

# Define and estimate a model
fit <- recent_production |>
  model(SNAIVE(Beer))

# Look at the residuals
fit |> gg_tsresiduals()
```



```
# Look at some forecasts
fit |> forecast() |> autoplot(recent_production) +
  labs(title="Forecast of Beer Production using Seasonal Naive")
```



What do you conclude?

The residuals appear to be roughly centered around zero and normally distributed indicating homoscedasticity. This is a good indicator that our model has explained much of the variance in the data and there is little to no “signal” left, only “noise”.

1.4 Exercise 5.4

Repeat the previous exercise using the Australian Exports series from `global_economy` and the Bricks series from `aus_production`. Use whichever of `NAIVE()` or `SNAIVE()` is more appropriate in each case.

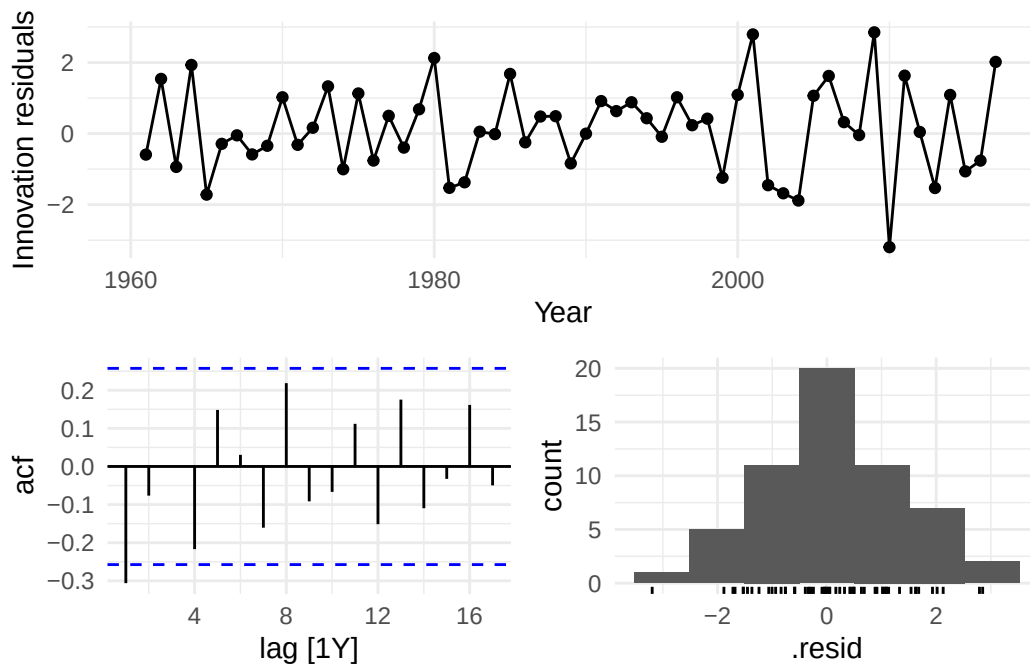
1.4.1 Australian Exports

Using a naive model for the Australian exports, we get a very well distributed set of residuals, normally distributed and a mean of zero. While the range of forecast values is wide, the naive method does a good job of extracting the available signal.

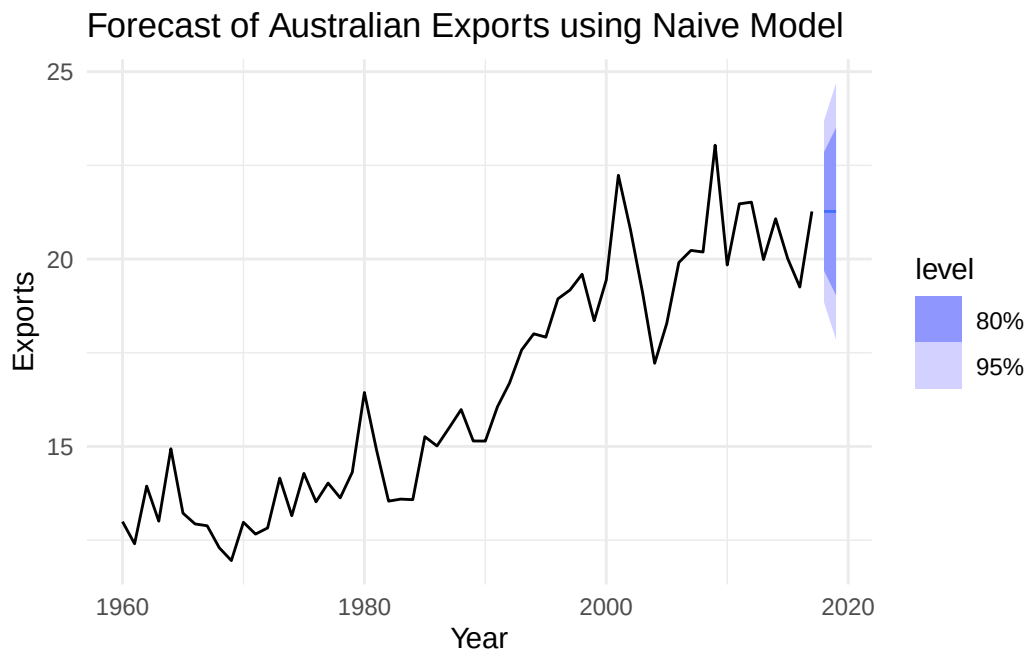
```
#
aus_exports <- tsibbledata::global_economy |>
  select(Year, Exports) |>
  filter(Country == "Australia")

# Define and estimate a model
aus_exports_fit <- aus_exports |>
  model(NAIVE())

# Look at the residuals
aus_exports_fit |> gg_tsresiduals()
```



```
# Look at some forecasts
aus_exports_fit |> forecast() |> autoplot(aus_exports) +
  labs(title="Forecast of Australian Exports using Naive Model")
```

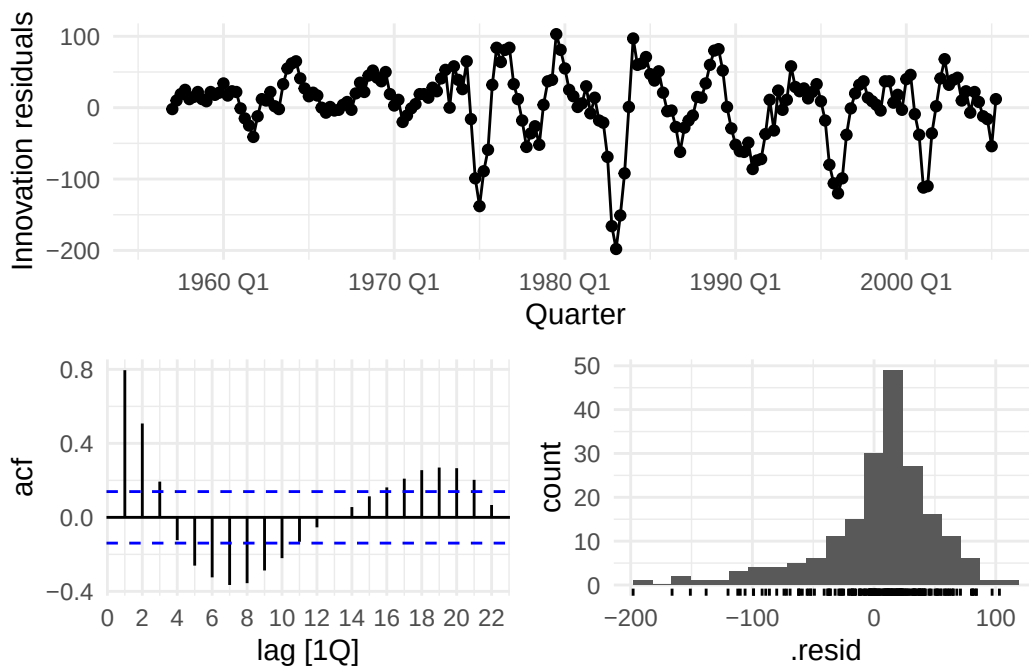


1.4.2 Bricks

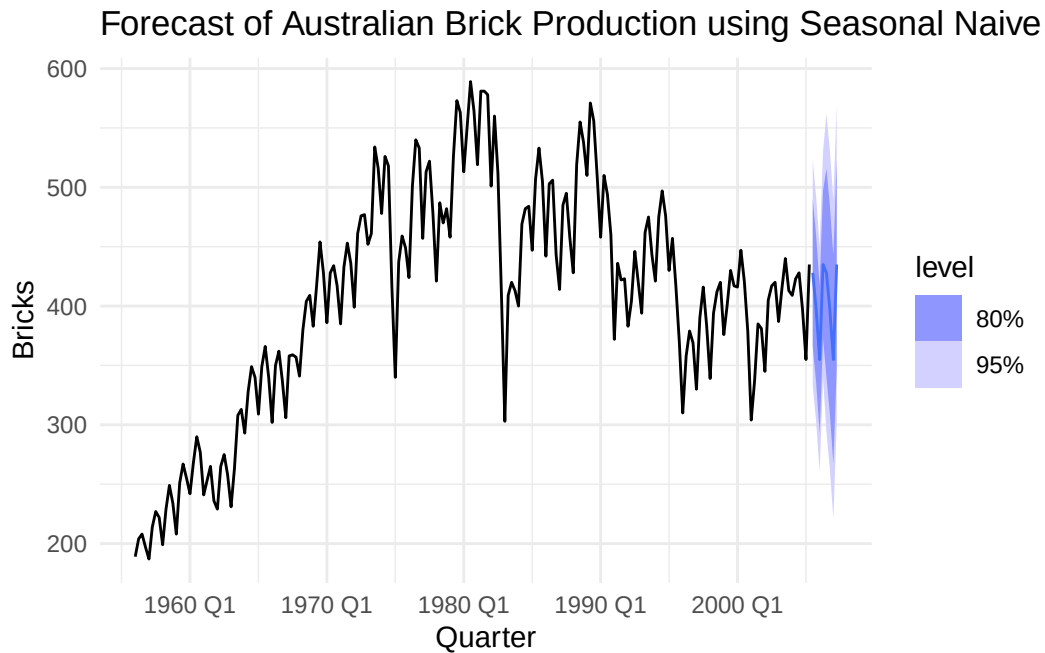
Neither naive (not pictured) nor seasonal naive seem to do particularly well for the brick dataset. While technically our residuals are centered around zero, the ACF and residual plot indicates to me there is some additional level of seasonality that this naive model is failing to capture. Seeing the ACF cycle and the patterns in the residuals plot, I would be confident that better forecasting performance could be achieved with a model that could better capture intra-season nuances.

```
# Define and estimate a model
bricks_fit <- bricks |>
  model(SNAIVE())

# Look at the residuals
bricks_fit |> gg_tsresiduals()
```



```
# Look at some forecasts
bricks_fit |> forecast() |> autoplot(bricks) +
  labs(title="Forecast of Australian Brick Production using Seasonal Naive")
```



1.5 Exercise 5.7

```
# Initiate "myseries" from prior exercise
set.seed(42)

myseries <- aus_retail |>
  filter(`Series ID` == sample(aus_retail$`Series ID`,1))
```

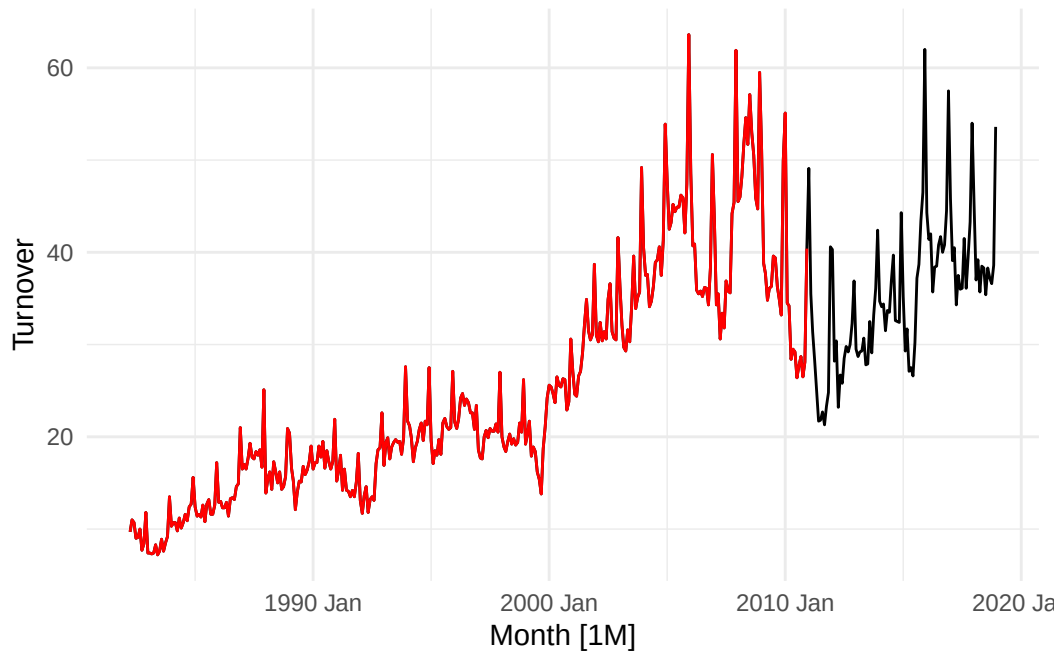
For your retail time series (from Exercise 7 in Section 2.10):

1. Create a training dataset consisting of observations before 2011 using

```
myseries_train <- myseries |> filter(year(Month) < 2011)
```

2. Check that your data have been split appropriately by producing the following plot.

```
autoplot(myseries, Turnover) +
  autolayer(myseries_train, Turnover, colour = "red")
```

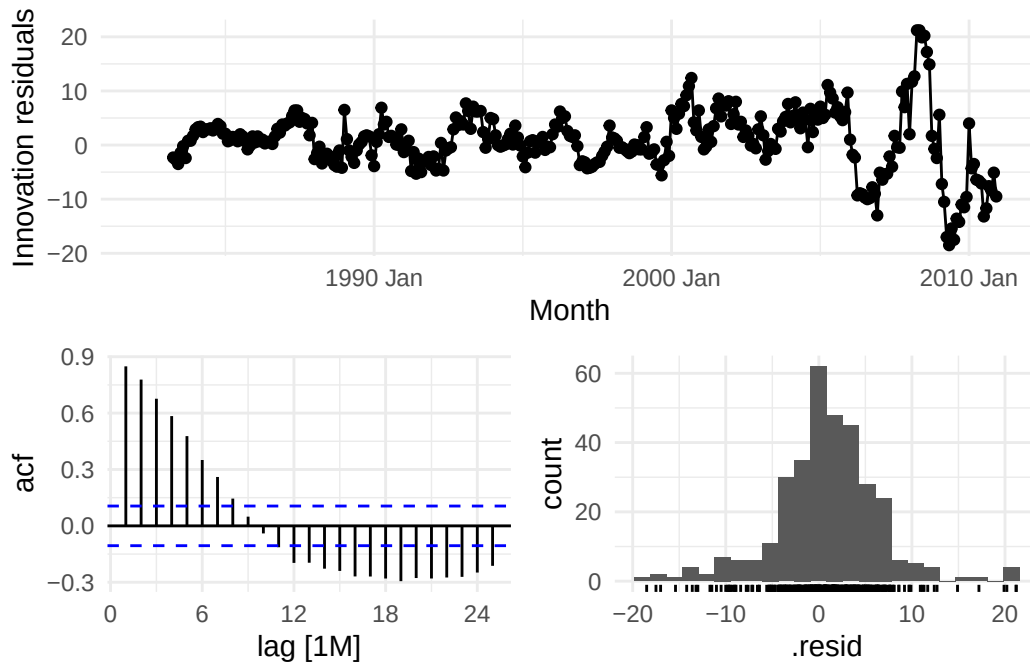


3. Fit a seasonal naïve model using `SNAIVE()` applied to your training data (`myseries_train`).

```
fit <- myseries_train |> model(SNAIVE(Turnover))
```

4. Check the residuals.

```
fit |> gg_tsresiduals()
```



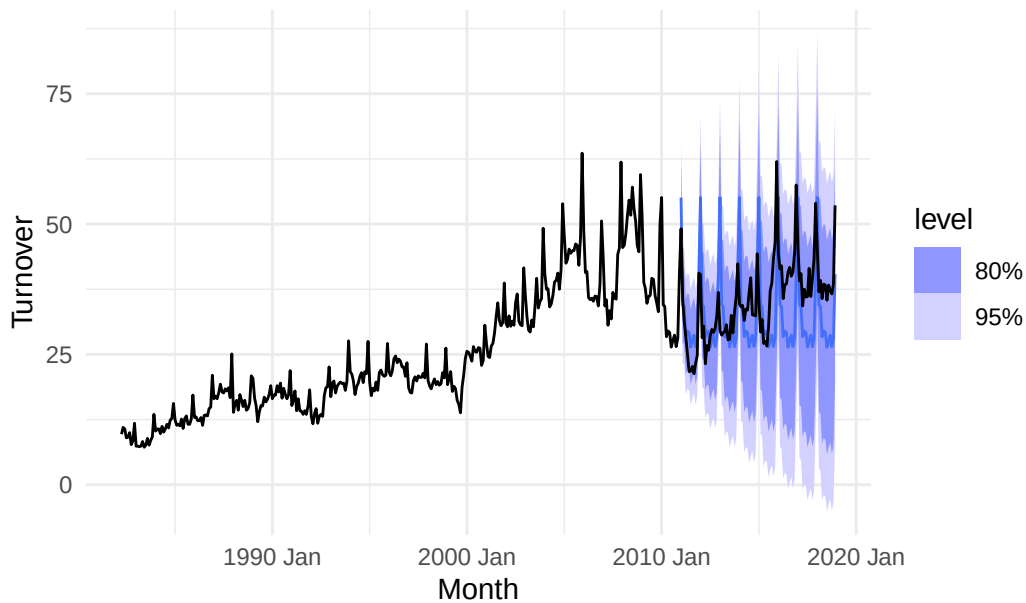
Do the residuals appear to be uncorrelated and normally distributed?

The residuals are normally distributed, however they are not uncorrelated - the ACF graph shows that very few months are within statistical expectations for random correlation, which tells us there is more signal to be uncovered in this data.

5. Produce forecasts for the test data

```
fc <- fit |> forecast(new_data = anti_join(myseries, myseries_train))
fc |> autoplot(myseries) + labs(title="Forecast of Retail Turnover")
```

Forecast of Retail Turnover



6. Compare the accuracy of your forecasts against the actual values.

```
fit |> accuracy()
```

```
# A tibble: 1 x 12
  State      Industry .model .type    ME  RMSE  MAE  MPE  MAPE  MASE  RMSSE  ACF1
<chr>    <chr>    <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 Western~ Newspaper~ SNAIV~ Trai~ 0.838  5.67  4.09  2.49  16.0    1    1  0.849
```

```
fc |> accuracy(myseries)
```

```
# A tibble: 1 x 12
  .model      State Industry .type    ME  RMSE  MAE  MPE  MAPE  MASE  RMSSE  ACF1
<chr>    <chr> <chr>    <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 SNAIVE(T~ West~ Newspaper~ Test   3.04  9.11  7.34  6.06  19.9  1.79  1.61  0.513
```

7. How sensitive are the accuracy measures to the amount of training data used?

The testing data compared worse across all measures compared to the training data, which is expected. The in-sample training data that the model sees should in general be more accurate than future, “unseen” data. From the chart it appears that the seasonality and forecasts were within a reasonable range, however the lack of additional data from 2010 and means the model could not account for the upwards trend. As such, our actual forecast values are off the mark, with a $MASE = 1.79$.